

The Art of Memory and Learning

Free resources to help you learn how to memorize anything and develop any skill

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How to create more memorable images, deal with difficulties and improve your skills

About this page: Once you understand the basics of memorization techniques, this page will help you push your skills a few steps further. Ideally, you should have already done at least one or two exercises with a memory palace. Although the majority of the examples presented will revolve around the memorization of lists of words, most of the strategies presented can just as well be used in other contexts. Even if you aren't interested in "memory sports" and you only care about preparing for your exams, I think you should still take the time to read and understand this page.

It's a lot I know! Especially if you click on all the links. You don't necessarily need to read everything. For now, you can focus on understanding all the main points while skipping over the links and various examples. The rest will still be there some other week if and when you will want to explore it.



A few important notes before we start:

* I will deliberately avoid choosing memorization examples where I can easily come up with some particularly memorable images or tricks that seem almost perfectly designed for the situation. I won't be presenting examples like: *"To remember Oliver's name, just imagine him eating olives."* Instead I'll be doing more or less the opposite. I'll choose difficult examples and I'll suggest memory tricks that are sometimes intentionally bad or mediocre. The main reason why is that I want to show you the extent of what is possible. I want to emphasize the fact that even the most difficult situations have possible solutions – and that even "bad" memory tricks can still be useful.

** I'll be presenting many different kinds of strategies you can choose to use. But to be clear: you certainly don't need to use ALL of them simultaneously for everything all the time. Use what you need, when you need it. Try to avoid overcomplicating the process for no good reason.

***The strategies explained on this page will be most useful when you are trying to memorize something that is particularly difficult to remember. In those cases, using the art of memory will almost always be your best bet. In other circumstances, if you are learning to use a new software or you want to understand how phenomenon X interacts with thing Y, this will not always be the case. For many real-world subjects and circumstances, it is often possible to remember something by paying close attention, by understanding its internal logic, by connecting in some way what you are learning with what you already know, and by frequently testing yourself in different ways. There's nothing wrong with using those more direct approaches when you have the time and energy and opportunity to do so. Learners can choose to use memorization techniques when they seem useful or necessary or when they simply feel like it. The rest of the time, they may choose to ignore them or to use them in a more limited way, for what's harder to remember. What's "harder to remember" can be very little or a whole lot depending on the subject and depending on your goals.

[I'm working on a complementary article about how to use the art of memory for complex subjects. Link will be added as soon as it's ready.]

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Some suggestions for dealing with more difficult words or names or concepts

It's great if you can find a way to directly represent what you want to remember. To remember the words "president" and "horse", you can just imagine Biden or Trump or Obama or Bush or whoever riding a horse. You can choose to add something to this image to make it more memorable, but here it's probably not even necessary. Okay so that was ridiculously easy, but you will often be facing situations that aren't so simple. Thankfully, there are many other strategies that are available and that you can experiment with.

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Learn to make the abstract more memorable

While concrete nouns are usually easy to visualize, you'll need to be more creative when dealing with more abstract concepts. If you have to remember a word like "tomorrow", you could use

either a calendar or some pile of boring paperwork that you would rather not have to deal with today. For “possible”, you might think of some insane athletic feat or some weird futuristic sci-fi type of object that you would look at and wonder whether or not it’s possible. Or you could picture some models “posing” (not the same word but close enough) for a picture or fashion parade. For “energy” you could imagine yourself doing push-ups or think of the energizer bunny. For “spring” you could just think of a tree with lots of leaves. Or you might imagine some people celebrating spring break in a way that I’m not going to describe here...

For a word like “grammar”, you can use a difficult sentence being written on the wall. You can use a teacher explaining grammar. You can use your grandmother (“grandma”...grammar...get it?), a “grandmaster”, a “grand mayor” or a “Grammy” award-winning artist. But let’s pretend for a moment you don’t have any inspiration and that no good option comes to mind. In those cases, it will always be possible to at least vaguely represent only one sound or even just one letter. For the very first letter of “grammar”, you could think of “golf”, a “game”, a “gorilla” or something or someone “gorgeous”. Or you could think of some “macaroni”, just for the “ma” in “grammar”. Some of those last few examples are really, really bad. But as we will see, a really bad trick is often better than nothing.

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Use any means necessary! And remember that even “bad” memory tricks can still be useful.

You’re trying to remember something difficult that has no easily understood internal logic and no self-evident memory trick comes to mind? It’s ok, you just need to try some unconventional tactics and go a little beyond the obvious. Pretend that you’re a particularly imaginative 7-year-old who isn’t shy about letting his or her mind go wild. Use whatever associations that come to mind, no matter how twisted, nonsensical or vague they may be. It doesn’t matter if your trick is only vaguely related to what you want to memorize.

- A good percentage of your tricks are likely to be based on a direct or indirect mental link between the meaning of the concept you want to remember and the image you are going to use to remember it. And when I say “indirect”, sometimes I mean “extremely indirect”. Maybe the word “incendiary” makes you think of a fire truck, a burning house or a crazy-looking person carrying a gas canister. Or perhaps it reminds you of someone expressing some “incendiary” argument about this or that controversial subject. Maybe the concept of “empiricism” brings to mind a cliché of a scientific experiment, with scientists wearing lab

coats and handling vials of colored liquid. Maybe it reminds you of some teacher, author or scientist. Or maybe it reminds you of that rich and pretentious student who gets humiliated by Matt Damon in a bar at the beginning of the movie Good Will Hunting. The word “empiricism” wasn’t even uttered in the scene in question, but it’s easy to imagine that it could have been.

- Another large portion of your tricks will be based on different words that sound somewhat similar, even if the meanings of those words are completely different. Let’s check some examples with the same two words we just mentioned. A word like “incendiary”, besides the more obvious associations, could also be represented by some incense, by a woman named Sandy, by an “innocent” prisoner, by an “incel”, by sand, by a diary or by some cheese and milk (dairies) or by countless other words or expressions. You could even use a dentist if it makes you think of those “incisive” teeth. “Empiricism” might remind you of Julius Caesar and his empire, an EMP weapon, some pirates or piranhas, a prism or some prices going up... Or even just some stupid rice representing the “ricism” part in “empiricism”? Not all of those tricks are equally good, but they can still be potentially useful. This remains true even when the sound of our “trick” is only very superficially similar to the sound of the word to remember.
- It’s ok as long as you can make some sort of mental association in your mind, even it doesn’t seem to make sense and you can’t explain it. Does the word “blandishment” make you think of Dracula for some mysterious reason? Using this image can work very well, even if it doesn’t seem to make any sense.
- In some much rarer cases, you may even choose to use some associations that are 100% arbitrary. If you have to memorize something difficult and no good or bad idea comes to mind, you could choose an image that you like almost completely at random. Encephalitis becomes a cockroach. Diphtheria becomes Elvis. Gastroenteritis becomes a Formula 1 car. I could have chosen other images with indirect links or vaguely similar sounds, but here I chose completely at random. This last method is difficult and far from ideal, but I think it can still have its place sometimes, especially when a tough concept comes up frequently. If you often have to memorize facts about encephalitis, being able to always represent this disease with a cockroach will be handy in the long run. Chances are that your brain will manage to remember that you thought of a cockroach when you started reading this tremendously exciting article on encephalitis. Even if you don’t get it right the first time, it will certainly be possible to create a more lasting mental association after a few more revisions.

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Whenever it seems useful or necessary, break things down into smaller pieces

If you have no idea how to represent a particular word, start breaking it up into parts. A word like “pulchritudinous” probably won’t evoke anything to you the first time you hear it, but what about shorter versions of that word like “pulch”, “ritu” and “dinous”? Could it become something like “push” (someone pushing something), “rite” (as in ritual, someone dancing around a fire maybe) and then “diner”? That crazy word could become someone who “pushes” (pulch) its “diner” (dinous) off the table as a form of “ritual” (ritu). Or maybe it could become you pushing around a “Ritudinous” (a just-made-up name for a dinosaur that is very into all kinds of rituals)? Those particular images are absurd and vague and stupid. The order isn’t quite right and it doesn’t sound exactly the same. But with a few trials and errors, I’m sure you can manage to use “pushes”+“ritual”+“diner” to remember “pulchritudinous”.

*When all else fails, as I said you should at least be able to find a trick for the very first letter of the word you’re trying to remember. Or if not for the first letter, for some semi-important syllable. Not great and not good either, but better than nothing.

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Use as many images and as many tricks as you need, but no more than that

- For my previous “pulchritudinous” example, I suggested some mnemonics for all the major parts of that difficult word. However, you should know that this isn’t always necessary. A mnemonic that covers just a fraction of what you’re trying to remember can often be more than enough. You could decide that “pulchritudinous” is simply a new kind of weird fruit punch. With some conscious effort and one or two additional reviews, that trick may very well be enough.
- If you’re memorizing a list of random words or random numbers, you can’t really afford to abstain from using memory techniques. At least not if you want to be quick and efficient. You don’t have to represent every single sound or every single letter, but you do need to find a way to remember every significant piece of data in the correct order. Sometimes if you’re lucky just one simple image will be enough to represent 2 or 3 words, but that’s the exception and not the rule.
- If you’re studying or reading or trying to remember something from a podcast or an article or a conversation, the rules aren’t quite the same. Unless you really need or want to know something word for word, you usually shouldn’t try to memorize everything! Instead you should identify what you want to remember and use memory techniques only to the extent that it seems fun or useful or necessary.

Remember that not every memory trick needs to be visual

Vision is probably our most powerful sense, but it's not our only tool. All other kinds of associations can work, as long as they make sense in some weird way in your mind. If you meet someone named Neil and you think "that guy wants to invent something *new*" (new... Neil... close enough), that's not visual and maybe the guy has no desire to invent anything, but it can still work.

Say it out loud or say it in your mind, as often as you need

This method isn't always necessary, but in some cases it's almost unavoidable. "Pulchritudinous" is not an easy word, even with those memory tricks I just shared. You will probably need to practice saying it out loud at least once. The memory tricks are here to help, but they can't always do 100% of the work for you. Saying it out loud will help you remember all those tiny details (that L in the middle of the "pulch" part for example). Saying something is inherently more memorable because it's an action that you're actively doing. You're moving your lips in a different way with each syllable. Maybe you can hear yourself saying it. If you're in the library or you're around people or you just don't feel like making any noise, you can still say the word in your mind and silently move your lips as if you were talking out loud.

Some suggestions for making your memory tricks more memorable

Those suggestions are valid in all kinds of circumstances, but particularly when you're using a memory palace. When you're busy placing images inside a memory palace, it can often be enough to simply say "all right there's this thing over there and there's this other thing nearby". But although this simple memorization strategy will "work" most of the time, it's not ideal and you will probably end up needing to review more often. Your images and your stories will be much more memorable if you use some combinations of the tricks suggested below.

Go extreme!

Whenever you feel like it, don't hesitate to make your images wilder, weirder, uglier, more beautiful, more violent, smellier, more disgusting, with more tentacles, more ice cream and more blood! In your imagination, there's no such thing as "too extreme" or "too offensive". Don't be afraid to use images that your teachers or your parents would find to be shocking.

*That's of course all up to you. More mundane or cute or ordinary images can also work quite well. Some of the most highly skilled "memory athletes" that we know of aren't commonly using hardcore images. They prefer simpler and more conventional images with some kind of logical link to what they're trying to memorize.

*Friendly warning: While using extreme imagery can be fun and it can be useful, it probably shouldn't be something you're always doing all the time. If everything is always exploding and everyone is always naked, those "extreme" events won't stand out in your mind nearly as well.

Use your 5 senses

You can add sounds or even olfactory, tactile and gustatory elements to your images. Sometimes I'll even add some catchy song.

Use the power of stories

When inspiration strikes, try to think of a bogus reason why all this nonsense happens. Is this character trying to impress this other character? Did someone forget that microwave in the forest? Why did the dragon eat that bicycle? Is it because there was nothing better around? Is it because it's a metal dragon? Is it because he read in Cosmopolitan magazine that iron is good for health? The "logical" links (quotation marks are important here) thus created can greatly solidify your stories. ***Of all the tricks mentioned on that page, I think this is one of the most powerful.*** If you find something that can sort of stick and sort of "make sense" in a crazy and absurd and weird way, it will be much harder to forget. The only problem with this strategy is that it's often difficult to put into practice. And it requires more mental energy. Quite often, no

good ideas will come to mind. If you can't think of any bogus narrative element that could help make sense of what's happening, it's ok, just move on. You can still remember things in other ways.

* Another similar trick I often use is to try to imagine how would a character feel if he or she ended up in that situation. *"My mother would totally punch that vampire to protect me!" / "Trump would be so disgusted and humiliated if he had to eat that meal." / "This dog seems a little confused and not sure how to feel. Is the bear that just arrived a friend or a foe? He seems friendly and fluffy, but also kind of scary..."*

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Link your images to one another and make them interact

Making your images interact with one another will make them stronger and more memorable. That way you can often remember something you forgot by thinking about the image that came right before or right after. Sometimes remembering just one tiny detail will help reactivate a much larger network of images and information. An image can interact with another in all kinds of ways. It can bump into it, fall on it or run away from it. It can be thrown. It can transform. Image 1 can find a way to use image 2 in any way that you can think of. Don't always default to having one thing accidentally hitting another and then another and then another. Try to vary your approach as often as possible. Also, if you can use this technique and combine it with some kind of story elements (why is that happening?) as recommended above, you'll be nearly unstoppable.

If you're placing 2 or more images per location, it should go without saying that image 1 "needs" to be interacting with image 2 in some way. But it's also possible to link images that are further apart. A character can walk from one location to the next. Objects can be thrown. Shots can be fired. Words can be exchanged. Noise can be made. A magical portal can make a long-distance punch possible. The interactions can also be much more subtle. A character can for example simply notice what is happening next door and find it very strange. Or maybe it can smell it or have some kind of emotional reaction to it?

In a way, having your images interact is allowing you to combine the memory palace method with the linking method, by making sure that one element will always be leading you to the next. Images can be linked in all kinds of ways. As I just explained, the link can take the form of a direct and indirect form of interaction.

Sometimes, the linking process doesn't have to involve any kind of direct interaction between images. It can be just conceptual or pseudo-logical. It can be a simple thought in your head related to the differences or similarities between what's happening over there and what's happening in the other room. *"There sure is a lot of shooting happening in that room." / "Too bad that Obelix in the bedroom can't see that giant cake on the roof" / "That puppy is lucky to be far enough apart from that troll." / "A spoon is a really odd choice of tool for that task!"*

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Make your images interact with their environment

When using a memory palace, it's usually not a bad idea to have your images be linked to their surroundings in different ways. An image placed next to a window can receive sunlight. It can notice something happening outside. Or it can break that stupid window in a fit of anger. If the image is on the table, it can be leaving footmarks on it, it can stand on the edge or fall over or step up all over your dinner. If it's near the fireplace, it can of course catch on fire. Or it can feel the heat and use it to get a nice tan. Or it can admire the fire while thinking profound thoughts about the meaning of life.

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Use "strategic micro-reviews"

Here's one simple tactic that I suspect is widely used, but that I've never seen clearly explained before. "Strategic micro-reviews" is a semi-ridiculous expression I came up with to describe this tactic.

Suppose you just placed 2 images on a table and 2 images on some other piece of furniture nearby. Doing a "micro-review" here would mean taking just a second or less to ask yourself "What did I just imagine here?" and/or "What are the words that I'm trying to memorize here?". Same process if you just learned a new name. I see this as a very quick and limited form of retrieval practice. Since you created the images just a moment ago, you should be able to answer in no time. And you shouldn't have to check the answers. If you can't do that, it means you did something wrong. Using "micro-reviews" might slow you down a little at first, but it will also help solidify your newly improvised images. I think it's possible to learn to use this strategy in a very fast and efficient manner.

That's the gist of it. If you're curious, I added a few additional examples and explanations about micro-reviews in [this short article](#).

Some suggestions for possible reviewing strategies you can use while practicing

It should go without saying, but of course "micro-reviews" won't be enough if you're aiming for more significant amounts of data. Different reviewing strategies will be necessary depending on your goals, your abilities and the nature of the tasks. Some trials and errors will most likely be necessary here.

You can read [this blog post](#) for various details regarding how to optimize the process.

The basics can be summarized with this example. Here's one possible strategy that one could use to memorize a list of 70 words:

1. Make up images for the first 20 words and review them once.
2. Make up images for the following 20 words and review them once.
3. Make up images for the following 20 words and review them once.
4. Review everything that you've memorized so far once or twice.
5. During the last few seconds, make up images for the next 10 words and don't review them.
6. Turn the sheet over and prepare for recall.

And then during recall:

1. First write down the last 10 words that you memorised a moment ago but never reviewed.
2. Drop the pen and see if you can quickly go through the rest of your memory palace. If there's a gap somewhere, don't stress and don't waste time on it for now. The goal here is to get one last quick mental review before you risk forgetting anything.
3. Quickly write down everything you can, without getting stuck on any missing gap.
4. If necessary, try to find the missing images for those few missing words. Try to think of various elements that could help reactivate those currently missing parts. Was the image interacting

with the window? Was it fleeing from the nearby monster? Would going through the alphabet in your mind help you find the beginning of the word? Use anything that can help.

5. If you're not "just practicing" and you really care about whether or not you got everything right, take the time to double and tripple check everything. Otherwise, just don't bother :)
6. Congratulate yourself for having perfectly or almost perfectly memorized a list of 70 words in order in just a few minutes!

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Some suggestions to improve a little faster

Once you've understood all the basic strategies, [here's how you can push things to the next level](#). You don't need to read this other page now. Complete or near complete beginners should probably wait a little more before exploring it. But I do think most people should come back to it at some point soon enough.